

Doña Ana County Water Quality — Monthly Sampling SOP Notes

Document

Working notes on the standard operating procedure for the 12-site monthly water-quality sampling program, Doña Ana County Environmental Services. Maintained by Ian Salazar; reviewed annually with Dr. Samuel Olivera.

Sites covered

12 monitoring sites across the Mesilla Valley: Picacho Reach upstream, Mesilla Diversion Dam, Las Cruces Wastewater outfall, La Mesa drain, Vado canal, Anthony NM agricultural drain (upstream cluster, sampled second Wednesday); Sunland Park downstream, El Paso confluence, American Dam, Riverside drain, Old Mesilla bridge, Hatch upstream (downstream cluster, sampled last Wednesday). Two-Wednesday cadence preserves comparability with prior years.

Field protocol — pre-departure

Equipment check list, night before. Calibrate the YSI multiparameter probe with fresh standards (pH 4.0, 7.0, 10.0; conductivity 1413 $\mu\text{S}/\text{cm}$). Confirm sample bottles labelled with site ID, date, collector. Confirm cooler ice packs frozen. Confirm Garmin GPS battery and coordinates loaded. Confirm chain-of-custody log book and EPA Region 6 sample tags. Coffee thermos. Scout's water bowl.

Field protocol — at site

Three readings per parameter (pH, DO, turbidity, temperature, conductivity). Discard the first; record the next two. Collect grab samples for laboratory analysis: arsenic, lead, copper (acidified, HNO_3), total coliform (sterile, no preservative, on ice). Photograph the sampling point with the Nikon DSLR — same composition each visit for time-series comparison.

Chain of custody

Sample bottles into cooler, locked, signed transfer log. Direct delivery to the county lab same afternoon for coliform (24-hour incubation window). Metals samples transferred to Ridgmont State's lab via courier within 48 hours, per the MOU with Dr. Brennan.

Reference values

EPA Region 6 thresholds: arsenic 10 $\mu\text{g}/\text{L}$, lead 15 $\mu\text{g}/\text{L}$, copper 1,300 $\mu\text{g}/\text{L}$ (action level), total coliform <1 CFU/100 mL (drinking water; receiving waters use a separate framework). Field readings flagged red in the Airtable tracker if any single sample exceeds threshold; cross-checked against laboratory confirmation before reporting up to Dr. Olivera.